

19th May 2025

MIT MORADABAD

1st CT

Department of Applied Sciences & Humanities

Course: B. Tech.

Semester: 2nd

Session: 2024 - 25

Sections: A, B, C, D

Subject: Physics

Subject Code: BAS 201

MM: 20

Time: 1 Hr

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Q. No.	1	2	3	4	5	6
CC: Blm	1 K3	4 K3	3 K2	3 K2	3 K2	4 K2

PART A

1.	Kinetic Energy of an electron is 4.55×10^{-25} J. Calculate momentum and wavelength of electron.	2
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PART B

2.	Calculate the numerical aperture, acceptance angle and critical angle of a fibre having core refractive index 1.466 and cladding refractive index 1.446.	3
3.	In Newton's ring experiment, the diameters of 4 th and 12 th dark rings are 0.400 cm and 0.700 cm respectively. Deduce the diameter of 22 nd dark ring.	3

PART C

4.	Discuss the phenomenon of Fraunhofer diffraction at a single slit and show that the relative intensities of the successive maxima are nearly $1 : (4/9) \pi^2 : (4/25) \pi^2 : (4/49) \pi^2 : \dots$	4
5.	Discuss the phenomenon of interference of light due to thin films in reflected light and find the conditions for maxima and minima.	4
6.	Describe construction and working of Ruby Laser with neat and labeled diagrams.	4

OR

Explain briefly Absorption and stimulated emission.
Establish a relation between Einstein's A and B coefficient.

$$R_{12} = R_{21}(Sp) + R_{12}(St)$$

- Write down proper theory wherever necessary.
- Draw Neat & Labeled Diagrams using Pencil only.

Lalit

[Signature]

20th Jun 2025

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2nd CT

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Q. No.	1	2	3	4	5	6
CO/ Blm	5 K2	2 K3	1 K3	5 K3	2 K4	1 K2

PART A

1.	Write a short note on Quantum Dots.	2
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PART B

2.	If the earth receives $3 \text{ cal min}^{-1} \text{ cm}^{-2}$ solar energy, what are the amplitude of electric and magnetic field of radiation?	3
3.	An electron is bound in 1-D potential box which has width $2.8 \times 10^{-10} \text{ m}$. Assuming the height of box to be infinite, calculate the lowest 3 permitted energy values of electron.	3

PART C

4.	Explain Type I and Type II superconductors? A superconducting tin has critical temperature of 3.7K in zero magnetic fields and a critical field of 0.306 T at 0 K. Calculate the critical field at 2 K.	4
5.	Write down Maxwell's equations for free space and prove that the electromagnetic waves are transverse in nature.	4
6.	What do you understand physical significance of wave function. Derive Schrodinger Time independent and Time dependent wave equations. OR Explain Compton Effect and derive an expression for the Compton shift. X-ray of wavelength $0.015 \text{ \AA}$ is scattered at 60° . Find the wavelength of scattered X-ray.	4

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18.06.2025